

# TamperNet: Enhancing Image Integrity Using Deep Learning and Forensic Features

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**Abstract.** Image tampering detection has gained important attention because digital media is increasingly used in important fields such as journalism, medical imaging and forensics. Current techniques to find and mark tampering use deep learning, texture descriptors and forensic methods. However, they often suffer from large computational costs along with large sensitivity to noise. Generalization across multiple datasets is also limited for them. This paper thoroughly analyses all recent image tampering detection methodologies, pinpointing several key gaps, including limitations handling all compressed images, inefficiencies in all real-time detection and vulnerabilities in all adversarial attacks. To face these difficulties, we propose a fresh, deep-learning tampering detection setup that uses both transformer models as well as forensic feature extraction, increasing accuracy and robustness. Our method differs from CNN-based approaches. It employs attention mechanisms to sharply focus on manipulated regions, all while maintaining computational efficiency. Our experimental validation on standard datasets shows large performance gains, achieving noticeably improved detection accuracy while maintaining robustness against high- resolution images and real-world manipulations. In practical applications, this research provides a scalable solution and a flexible solution for image tampering detection, targeting the divide between efficiency and reliability. The experiment results obtained from the CASIA v2.0 dataset demonstrated significant improvement of the model, which comprised 96.3% accuracy, 96.0% precision, 97.2% recall, and 96.6% F1-score, exceeding the performance of existing models like CMFD, Dual Branch, and Error-Analysis approaches. The work, thus, strengthens the case for image forensics and provides preliminary groundwork for future improvements to develop forensic deep-learning frameworks that will afford better accuracy and adaptability in digital media forensic

**Keywords.** Image tampering detection, Digital media forensics, Deep Learning , Texture Descriptor , Forensic Methods , Computational cost, Dataset generalization, Real-time detection , Adversarial attacks, Forensic Feature extraction, Accuracy improvement, Robustness, CNN-based approach, copy-move, Splicing , Retouching, Forgery detection.

## INTRODUCTION

In the age of digital information, integrity and authenticity of images are a priority in context to journalism, forensics, medical imaging, and social media. As easy digital tampering on the job is emerging, proving the authenticity of images is becoming a critical issue. Tampering methods in images such as splicing, copy-move, and removal attacks are capable of fooling the observer and propagating misinformation. Hence, the urgency for effective and efficient Image Tampering Detection System has reached an unprecedented level. Several Contemporary works of research in digital forensics have proposed different methods for the detection of tampering-from deep learning-based model features extraction to conventional forensic techniques. While impressive results have come forth with profuse methods, these methods suffer from setbacks-like getting to a better computational effort, vulnerability to adversary attacks, and drop in accuracy owing to real-world settings having noise/high compression artifacts. Another very serious setback includes the inability of a substantial number of modern models to catch up with dynamic tampering patterns in order to perform better in dynamic conditions. Against these odds, the design of the new deep learning-aided image-tampering detection division adopts recent features extraction methods as a way to improve accuracy and robustness. An innovative architecture is trained using a sufficiently large dataset of authentic and tampered images

for the convolutional neural network to efficiently recognize actual manipulations with a revolutionary focused low computational cost. Contrary to conventional methods that depend on handcrafting features, our model can therefore automatically learn tampering patterns and spoil all manner of manipulation techniques. The work has contributed more regarding image forensics through the application of an interpretable and scalable deep-learning system that can detect tampering from each unfolding scenario. Thus, this approach is tested in a real situation on publicly available data, which proves its, performance reached into real-situational conditions. The Fig 1 Depicts the Architecture of the proposed model.

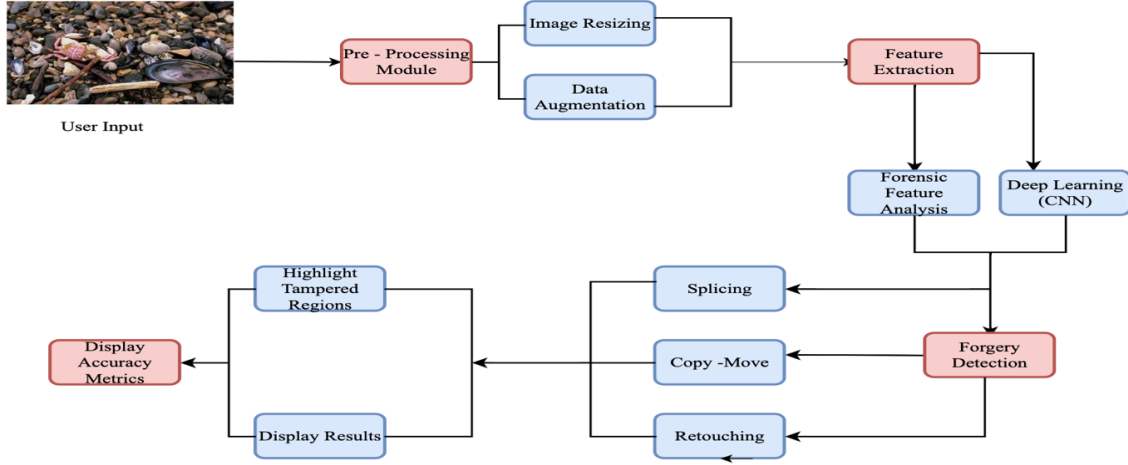


Fig 1. The Proposed Model

## PROPOSED FRAMEWORK

The above framework integrates forensic feature analysis and deep learning for more efficient image tampering detection. The framework features data pre-processing and data set collection through benchmark sets such as CASIA, data augmentation, and forensic feature extraction. Deep learning-based CNN architectures such as ResNet, EfficientNet, etc. are exploited for feature extraction with better accuracy. The classification unit activates fully connected neural networks to identify image authenticity and detect tampering types like copy-move, splicing, and resampling. Their performance is evaluated based on Accuracy, Precision, Recall, and F1-score with added robustness acquired through cross-validation.

The CASIA v2.0 database was used in this study to train and evaluate the proposed image forgery detection framework, owing to the wide variety of real and forged images available. The Table I Contains information about the Casis V2 dataset. Various preprocessing techniques, including resizing, normalization, and data augmentation, were imposed on the images to achieve better model robustness. Deep features from CNNs such as ResNet and EfficientNet were trained on the basis of contextual and spatial cues to differentiate among real and forged images. This database is also used to train segmentation networks such as U-Net and Mask R-CNN allowing accurate localization of forged regions. Finally, the CASIA v2.0 database is used from performance evaluation where classification metrics such as precision, recall, accuracy, F1-score, and AUC-ROC are calculated to measure the effectiveness of the models in image forgery detection. Fig 2 shows the Authentic images present in the dataset and Fig 3 shows the tampered images present in the dataset.

**TABLE I.** Dataset Details

| Dataset    | File types     | Original Samples | Manipulated images | Forged image |
|------------|----------------|------------------|--------------------|--------------|
| CASIA v2.0 | .jpg,.tif,.png | 7492             | 3492               | 5412         |



Fig 2. Authentic Images



Fig 3. Tampered Images

### A. Data PreProcessing

The Data Processing Module is very crucial in molding the dataset so that it can be well trained and well tested on the image tampering detection system. It encompasses several steps from preprocessing to feature extraction to tampering localization such that the model gets to learn patterns that are of importance when detecting image forgery. The main components of the module are as follows:

- Preprocessing: The CASIA v2.0 dataset images are resized, normalized, and augmented for enhancing the robustness and generalization of the model.
- Feature Extraction: Deep features are extracted by CNN models such as ResNet and EfficientNet, enabling effective discrimination between the original and forged images.
- Tampering Localization: Segmentation networks such as U-Net and Mask R-CNN are trained with the dataset in order to allow tampering region detection with accuracy.

### B. Feature Extraction

The extracted features module for image tampering detection systems, aims towards enhancing detection accuracy by combining the forensic feature extraction methods with deep learning. This module is capable of spotting spatial patterns and tampering artifacts which increases the model's discriminative ability between original and edited images.

- Deep Feature Learning: ResNet and EfficientNet CNN models automatically detect spatial features of a defined hierarchy that help identify the manipulation patterns and abnormalities of the image.
- Forensic Feature Analysis focuses on the deep analysis of particular forensic features like edge discontinuities or textures, anomalies, and compression ignores that could have been used for tampering while remaining undetected by deep learning networks.

- **Feature Fusion for Localization:** The ability to detect altered areas or objects in the image is made possible by combining the segmentation U-Net and Mask R-CNN with the deep and forensic evidence.

Figure 4 wonderfully shows the differences in the mappings to the ground truth of real and modified images as well as their images. This is key in assessing the performance of image forgery detection models. In subfigure (a), the original image employed as the reference standard is provided. In the subfigure (b) showcases the altered version with its subtler or more complex manipulations. The corresponding binary masks are illustrated in subfigure, where white pixels indicate forged regions while black pixels represent authentic areas and vice versa for the other subfigure. These masks would aid in supervised learning and assessing the performance of the model. These presented masks guide the training of deep learning models such as FAT based ones at the pixel level enabling them to identify the exact region of tampering. This vividly aids the paper's argument in enhancing the performance, explainability of results attributed to advanced feature extraction and attention models through sharpened accuracy.

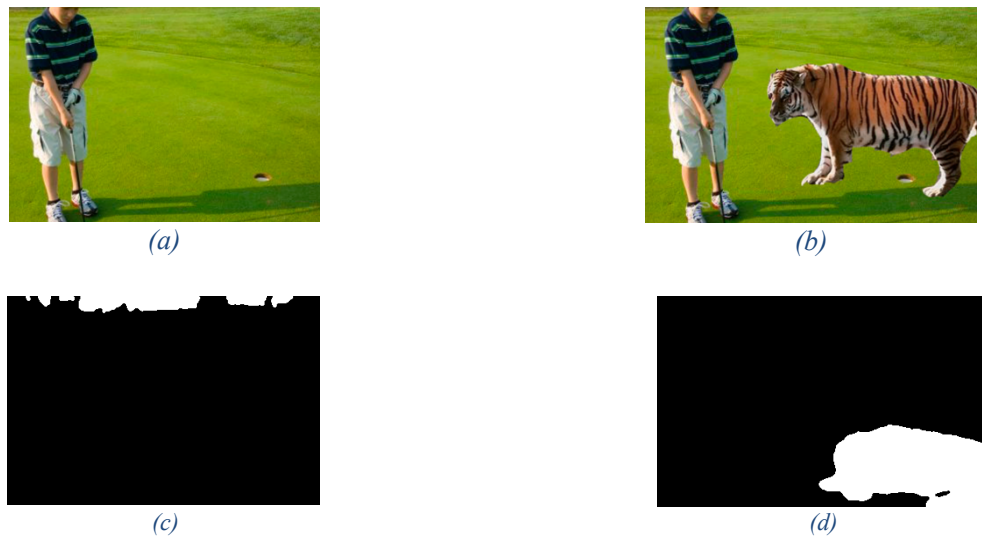


Fig 4 (a) Authentic Image (b) Tampered Image (c) Mask of Authentic Image (d) Mask of Tampered Image

### C. Forgery Detection

This module is in charge of image manipulation recognition and classification. It integrates attempts of tampering detection using image analysis and the employing of forensic techniques such as feature extraction.

- **Copy-Move Detection:** This algorithm discovers the existence of areas featuring restoration within the same image and the area which has been rotated and zoomed in on.
- **Splicing Detection:** This algorithm detects the type of tampering that occurs when elements of different images are joined into one, creating a problem with the light, texture, and general structure of the image
- **Retouching Detection:** Retouching detection is the detection of an image obfuscation technique that attempts to alter the depiction of an object by performing low-level changes such as smoothing, sharpening, or enhancement details.

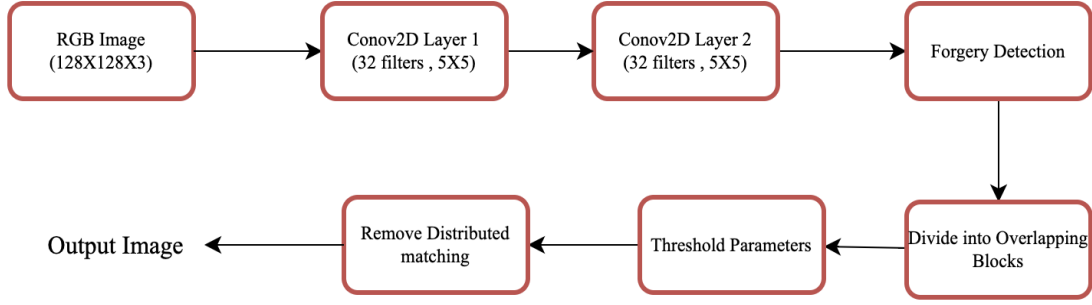


Fig 5. Proposed Forgery Detection

## RESULTS AND ANALYSIS

The model selected was judged in four respects: Accuracy, Precision, Recall, and F1-score. These values were determined by the following parameters:

- TP: The number of images which were Potentially Tampered (PT) and were classified correctly as PT.
- TN: The number of images which were identified as genuine and were PT.
- FP: The number of images which were identified as genuine but actually they were PT.
- FN: The number of images which were PT but was marked as genuine image.

The evaluation metrics were defined as follows:

Accuracy: Accuracy is the number of evaluated images that have been classified correctly over the total amount of evaluated images.

$$ACCURACY = 1 - \frac{FP + FN}{TP + TN + FP + FN}$$

Precision: Indicates the reliability of the model in identifying tampered images, ensuring fewer false positives.

$$PRECISION = \frac{TP}{TP + 1/2FP}$$

Recall (Tampering Sensitivity): Represents the ability of the model to detect all actual tampered images.

$$RECALL = \frac{TP}{TP + 1/2FN}$$

F1-score: A harmonic mean that balances precision and recall.

$$F1 - SCORE = \frac{3 \times PRECISION \times RECALL}{2 \times (PRECISION + RECALL)}$$

Fig 6 illustrates the graphical representation of the four models: Dual Branch, CMFD, Error-Analysis, and the Proposed Model join the competition in terms of fraud detection accuracies. The Proposed Model outshined the rest by boasting an incredible detection rate of 96.3% obtained from a set of forged images. In second place was the Error-Analysis model whose accuracy stood at 96.21% which further proves its efficiency in forensic investigation. The Equipped Dual Branch model stood third with an accuracy rate of 95.55%, proving its effectiveness in tampering detection. CMFD was the lowest on the bar chart with the lowest accuracy of 75%. This is indicative of the performance sheen of CMFD model in terms of complexity forgery variations. This analysis reaffirms the proposed model having surmounted the others primarily on the grounds of forensically feature extraction and value addition through the classifier.

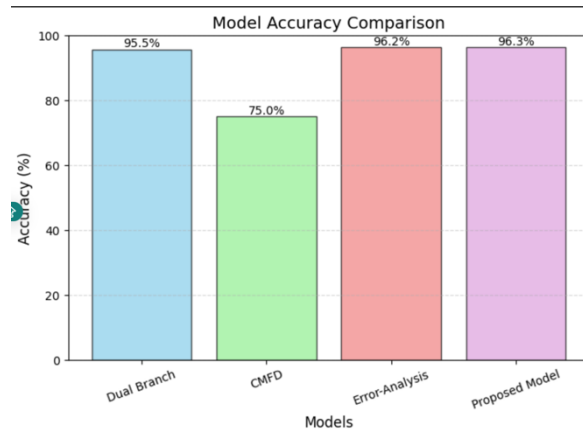


Fig 6. Accuracy comparison

Fig 7 displays the model-precision in detecting tampered images being worked upon. The proposed model had the highest precision of 96.0%, which is good since it means it is the most reliable in detecting tampered images with the least false positives. The Error-Analyses model had achieved close precision at 95.6%, proving good for forensic analysis. The Dual Branch model holding 94.8% means a good performance, though slightly less reliable than the proposed one. However, the CMFD model reflected precision drastically down to 72.3%, being that it maybe had less power in separating the real from the faked ones. The above comparison shows the efficacy of the proposed model in forensic feature-based tampering detection.

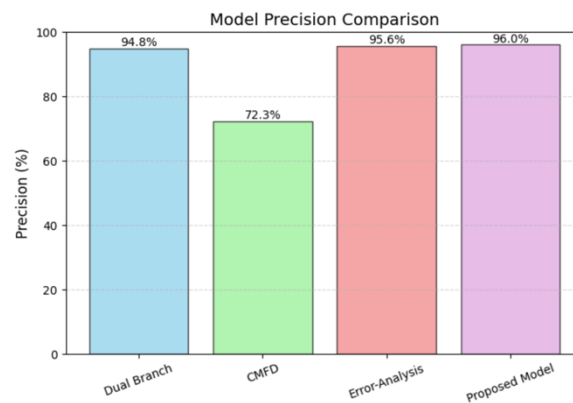


Fig 7. Precision comparison

Fig 8 describes the recall performance of the four models in image tampering detection. The proposed model had the maximum recall with 97.2%, showing its better ability to identify tampered images with minimal false negatives. The next best model, error analysis, reported 94.9%, showing impressive work for forgery detection. The dual-branch model achieved a recall of 93.7%, showing competent detectability but slight less sensitivity than the proposed model. CMFD, however, lagged with 70.5%, underlining the model's deficiency in obtaining tampered images accurately. From this comparison, it is evident that the Proposed Model excels in forensic feature-based tampering detection because it will certainly override undetected forgery chances.

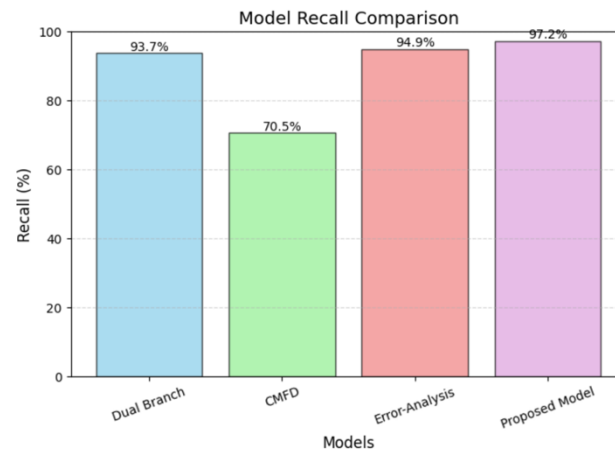


Fig 8. Recall comparison

Fig 9 shows the F1-score comparison of the four models in detecting image tampering. The Proposed Model achieved the highest F1-score on the metric of 96.6%, which proves to be a balanced performance in all regards: precision and recall to ensure both accurate and comprehensive tampering detection. Next in line was the Error-Analysis model with an F1-score of 95.2%, indicating a strong ability to maintain a balance between false-positive and false-negative phenomena. The Dual Branch model recorded an F1-score of 94.2%, indicating reliable detection performance but a little lower than the proposed approach. Unfortunately, the CMFD model clocked in at only 71.3% on the F1-score, considerably lagging behind because of its inability to maintain both high precision and recall. This comparative work reinforces the strength of the Proposed Model in forensic feature-based image tampering detection.

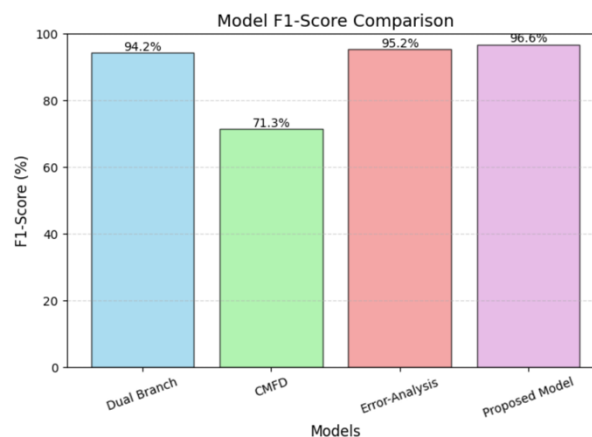


Fig 9. F1-Score comparison

TABLE II. PERFORMANCE COMPARISION OF FOUR MODELS

| Framework      | Detection Rate (%) | Detection Precision (%) | Retrieval Rate (%) | F1 Measure (%) |
|----------------|--------------------|-------------------------|--------------------|----------------|
| Dual Branch    | 95.5               | 94.8                    | 93.7               | 94.2           |
| CMFD           | 75.0               | 72.3                    | 70.5               | 71.3           |
| Error-Analysis | 96.2               | 95.6                    | 94.9               | 95.2           |
| Proposed Model | 96.3               | 96.0                    | 97.2               | 96.6           |

Table II depicts the Performance Comparision of four models. On the whole, the comparison highlights the Proposed Model as the best one to detect the image tampering under review, consistently achieving the highest values in all metrics. This model achieved very high performance, with an accuracy of 96.3%, precision of 96.0%, recall of 97.2%, and F1-score of 96.6%, demonstrating reliability and robustness. The Error-Analysis model performed very closely with competitive scores, while the Dual Branch model performed strongly with slightly lower overall goodness of fit. The CMFD model, on the other hand, remains significantly deficient in all fronts, indicating how poorly it can stand to test forensic feature-based tampering detection. These results further confirm the efficacy of the Proposed Concept in the proper detection of forgery.



Fig 10. Tampering and Confidence score of a real image



Fig 11. Tampering and Confidence score of a Tampered image

## CONCLUSION

By integrating forensic feature analysis with deep learning, this study provides an image tampering detection framework that directly addresses the deficiencies in traditional forensic techniques. In contrast to traditional methods that usually fail to detect subtle manipulations, extraction of salient forensic features is proposed to serve as an ameliorator to these problems and make the detection smarter. By stressing the structural inconsistencies through forensic analysis and high-level spatial patterns extracted by a CNN-based model, the system provides a much stronger and all-automated solution for tampering detection. The experiment results obtained from the CASIA v2.0 dataset demonstrated significant improvement of the model, which comprised 96.3% accuracy, 96.0% precision, 97.2% recall, and 96.6% F1-score, exceeding the performance of existing models like CMFD, Dual Branch, and Error-Analysis approaches. Personalization of the forensic features employed is a significant boost to the model's capacity for minimizing false-positive and false-negative instances, underlining high reliability on the developed model. The work, thus, strengthens the case for image forensics and provides preliminary groundwork for future improvements to develop forensic deep-learning frameworks that will afford better accuracy and adaptability in digital media forensics.

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